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Course: Diploma in Cyber Security

DAY 1 – OVERVIEW OF COMPUTER HARDWARE COMPONENTS

Topic: Overview of Hardware Components

INTRODUCTION TO COMPUTER HARDWARE

What is Computer Hardware?

Computer hardware means the physical parts of a computer that we can see and touch. For example, the motherboard, CPU, RAM, GPU, HDD, SSD, keyboard, and monitor are all hardware components.

A computer does not depend on one component alone. Different hardware parts have different jobs, but they work together like a team. For example, the CPU processes instructions, RAM keeps currently needed data ready, and storage keeps our files even after the computer is turned off.

Why is hardware important?

Hardware provides the physical foundation on which software operates. When we open a game, run a program, save a file, or browse the internet, several hardware components work together in the background.
Easy way to remember

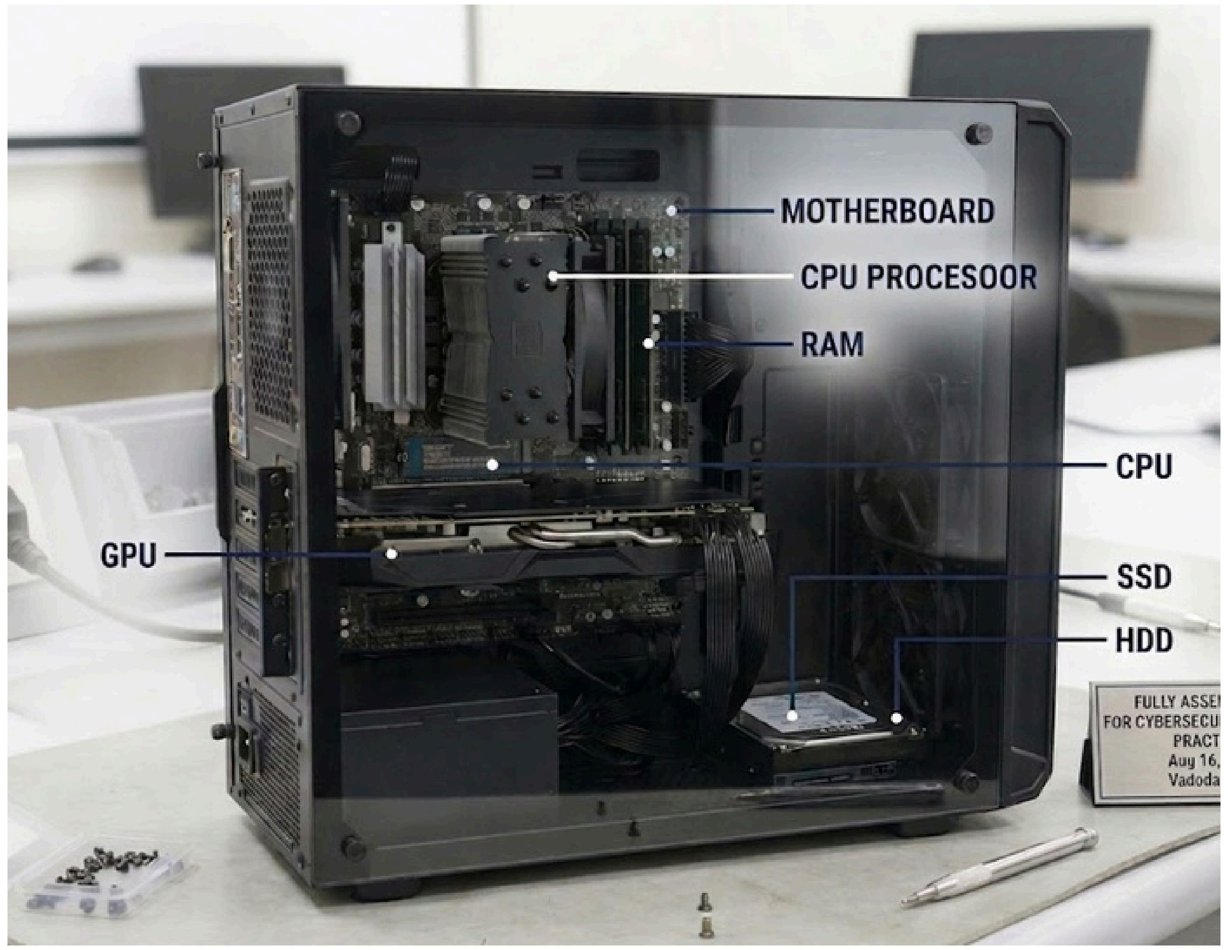
CPU = Thinks
RAM = Keeps things ready
GPU = Creates graphics
SSD/HDD = Stores things
Motherboard = Connects everything

LABELED DESKTOP PC DIAGRAM

Main Hardware Components

Components shown in the diagram

1. **Motherboard – Main communication platform of the PC.**
2. **CPU – Processes instructions and performs calculations.**
3. **RAM – Provides temporary working space for active programs.**
4. **GPU – Handles graphics and visual calculations.**
5. **HDD – Provides long-term storage using magnetic disks.**
6. **SSD – Provides faster long-term storage using flash memory.**



CPU, RAM AND MOTHERBOARD

CPU – Central Processing Unit

The CPU is the component that takes instructions, works on them, and produces results. It is responsible for a huge number of small decisions and calculations that happen while we use a computer.

If we open a calculator and enter 25 + 25, the CPU processes the instruction and helps produce the answer.

CPU = Processing

RAM – Random Access Memory

RAM is the computer's short-term working space. It temporarily holds the programs and data that the CPU needs to access quickly.

For example, when several applications are open at the same time, they use RAM. When the computer is switched off, the information stored in RAM is lost.

RAM = Temporary working space

MOTHERBOARD

The motherboard is the main circuit board that provides the connections through which the computer's major components communicate.

It contains or connects important parts such as the CPU socket, RAM slots, expansion slots, storage connections, and other interfaces.

Think of the motherboard as a road network. The CPU, RAM, GPU and storage are like different places connected by those roads

Motherboard = Connection platform

GPU AND STORAGE
GPU – GRAPHICS PROCESSING UNIT

A GPU is a processor designed to handle large amounts of visual and parallel calculations efficiently. It is especially important for games, 3D graphics, video editing, and other graphics-heavy applications.
For example, in a game, the GPU helps generate objects, lighting, textures, shadows, and other visual elements that appear on the screen.
GPU = Graphics processing

HDD – Hard Disk Drive
An HDD is a long-term storage device that saves data magnetically on spinning disks.
It can store operating systems, applications, photos, videos, documents, and other files. HDDs are generally cheaper per gigabyte but slower than SSDs.
HDD = Magnetic storage

SSD – Solid State Drive
An SSD is a storage device that saves data using flash memory instead of spinning disks.
Because an SSD has no moving mechanical disk, it can provide much faster access to data and is generally more resistant to physical shock than an HDD.

SSD = Fast flash storage

HDD VS SSD COMPARISON

Feature	HDD	SSD
FULL FORM	HARD DISK DRIVE	SOLID STATE DRIVE
TECHNOLOGY	MAGNETIC SPINNING DISKS	FLASH MEMORY
SPEED	GENERALLY SLOWER	GENERALLY FASTER
MOVING PARTS	YES	NO
SHOCK RESISTANCE	LOWER	HIGHER
NOISE	MAY PRODUCE MECHANICAL NOISE	SILENT
PRICE PER GB	USUALLY CHEAPER	USUALLY MORE EXPENSIVE
COMMON USE	LARGE, BUDGET-FRIENDLY STORAGE	OPERATING SYSTEM, APPLICATIONS AND FAST STORA

Which one should we choose?

If we want fast booting, quick application loading, and better responsiveness, an SSD is usually the better choice. If we need large amounts of storage at a lower cost, an HDD can still be useful.

HOW CPU, RAM AND GPU WORK IN A GAME

Example: PUBG or Free Fire on a PC

When we play a game, the computer does not use only the GPU. CPU, RAM, GPU and storage work together.

CPU – The decision maker

The CPU handles game instructions and calculations. It can manage things such as game logic, player actions, physics, and communication with other system components.

RAM – The active workspace

RAM holds the game and other currently needed information so that the CPU can access it quickly. Having enough RAM helps the system handle the game alongside other running programs.

GPU – The visual worker

The GPU processes graphics and helps generate the images shown on the monitor. It is particularly important for resolution, textures, lighting, shadows, and frame rendering.

Storage – The game cupboard

The SSD or HDD stores the game files permanently. When the game starts, required information is loaded from storage into RAM so the system can work with it.

Simple flow

Game stored on SSD/HDD

↓

Game data loaded into RAM

↓

CPU processes instructions

↓

GPU processes graphics

↓

Monitor displays the result

UPGRADING A PC FOR THE LATEST FIFA

If I want to run the latest FIFA smoothly
I would consider upgrading the following components:

1. GPU Why?

The GPU has a major effect on gaming graphics and rendering performance. A stronger GPU can help achieve better visual quality and smoother gameplay when the system meets the game's requirements.

2. RAM Why?

If the computer does not have enough RAM, the game and other running applications may compete for limited memory. Increasing RAM can give the system more working space.

3. SSD Why?

Replacing an older HDD with an SSD can significantly improve boot times and game loading times. It does not automatically increase FPS, but it can make the overall system feel much faster.

4. CPU Why?

The CPU handles game logic, calculations, and many background tasks. If the existing CPU is too weak for the game's requirements, upgrading it can improve overall gaming performance.

5. Cooling System Why?

Gaming creates more heat. Good cooling helps the CPU and GPU maintain suitable temperatures and can help prevent performance reduction caused by excessive heat.

Important point

The best upgrade depends on the existing PC. Upgrading one component does not automatically solve every performance problem.

CONCLUSION

What I Learned

Computer hardware is a combination of different physical components, and each component has a specific responsibility.

The CPU processes instructions, RAM provides temporary working space, and the GPU handles graphics processing.

The motherboard connects the major components, while HDD and SSD provide permanent storage.

Understanding these components is important not only for building and upgrading a PC, but also for troubleshooting computer problems and choosing suitable hardware for different tasks such as gaming, programming, and cybersecurity.